Applied Graph Theory

MAT 4409

Contents

List of Symbols

$V(G)$	Vertex set of graph G
$E(G)$	Edge set of graph G
$\sim$	is adjacent to
$\deg(v)$	Degree of vertex v
$\delta(G)$	Minimum degree of graph G
$\Delta(G)$	Maximum degree of graph G
$\mathrm{ecc}(v)$	Eccentricity of vertex v
$\mathrm{rad}(G)$	Radius of graph G
$\mathrm{diam}(G)$	Diameter of graph G
$A(G)$	Adjacency matrix of graph G
$B(G)$	Incidence matrix of graph G
K_n	Complete graph of order n (size $\binom{n}{2}$)
P_n	Path graph of order n (size $n-1$)
C_n	Cycle graph of order n (size n)
$L(G)$	Line graph of G

1 Algebraic and Structural Properties

1.1 Graphs and Degree

Throughout these notes, all graphs are finite, simple, and undirected unless stated otherwise.

A graph is an ordered pair $G = (V, E)$ where V is a non-empty set called the vertex set, whose elements are the vertices of G, and E is a set of unordered pairs of distinct vertices of G, whose elements are the edges of G. The order of G is its number of vertices and the size of G is its number of edges.

Two vertices u and v of G are adjacent, denoted by $u \sim v$, if $uv \in E$ (note that uv here denotes an unordered pair of vertices, and $uv = vu$). If there is no edge between u and v, then they are nonadjacent, denoted by $u \nsim v$. The vertex v and the edge uv are said to be incident with each other. The degree of a vertex v is the number of edges incident with it, or equivalently, the number of vertices adjacent with it, and is denoted by $\deg(v)$.

Lemma 1.1 (Handshaking Lemma). *The sum of the degrees of all vertices of a graph is twice its size.*

Proof. Let G be a graph. As each edge is incident with exactly two vertices, each edge is counted once by two terms in the sum of the vertex degrees. Thus,

$$\sum_{v \in V(G)} \deg v = 2|E(G)|. \qquad \square$$

Corollary 1.2. *The number of vertices of odd degree in a graph is always even.*

Proof. Exercise. $\square$

The minimum degree of G is $\delta(G) = \min_{v \in V(G)} \deg v$, and the maximum degree of G is $\Delta(G) = \max_{v \in V(G)} \deg v$. A regular graph is a graph in which all vertices have the same degree. This common value of the degree is the regularity of the graph. A regular graph of regularity k is a k-regular graph. A cubic graph is a 3-regular graph.

Exercise 1.1. Prove the following:

1. Any cubic graph has even order. For which other regular graphs will this statement hold?
2. In any graph on two or more vertices, at least two vertices have the same degree.
3. If G is a graph of order n and size m, then $\delta(G) \leq \frac{2m}{n} \leq \Delta(G)$.

1.2 Cartesian Products

The Cartesian product of two graphs G and H is the graph denoted by $G \times H$, whose vertex set is $V(G) \times V(H)$ (the Cartesian product of the vertex sets $V(G)$ and $V(H)$, consisting of all ordered pairs (u, v) where u is a vertex of G and v is a vertex of H), in which two vertices (u_1, v_1) and (u_2, v_2) are adjacent if and only if either $u_1 = u_2$ and $v_1 \sim v_2$ or $u_1 \sim u_2$ and $v_1 = v_2$.

Exercise 1.2. Prove that if G is a graph of order p_1 and size q_1, and H is a graph of order p_2 and size q_2, then $G \times H$ is a graph of order $p_1 p_2$ and size $p_1 q_2 + p_2 q_1$.

Solution. By definition, the vertex set of $G \times H$ is the Cartesian product $V(G) \times V(H)$ of the vertex sets of G and H, and therefore contains $|V(G)| \times |V(H)|$ elements. Thus, the order of $G \times H$ is $p_1 p_2$.

Now, consider any vertex (u, v) of $G \times H$. Corresponding to every vertex u' adjacent to u in G, there is an edge from (u, v) to (u', v) in $G \times H$. Similarly, corresponding to every vertex v' adjacent to v in H, there is an edge from (u, v) to (u, v') in $G \times H$. Thus, the degree of (u, v) in $G \times H$ is $\deg u + \deg v$. Hence, by the Handshaking Lemma, the size of

$G \times H$ is half the sum of its vertex degrees, which is

$$\begin{aligned}
|E(G \times H)| &= \frac{1}{2} \sum_{\substack{u \in V(G) \\ v \in V(H)}} (\deg u + \deg v) \\
&= \frac{1}{2} \sum_{\substack{u \in V(G) \\ v \in V(H)}} \deg u + \frac{1}{2} \sum_{\substack{u \in V(G) \\ v \in V(H)}} \deg v \\
&= p_2 \left(\frac{1}{2} \sum_{u \in V(G)} \deg u \right) + p_1 \left(\frac{1}{2} \sum_{v \in V(H)} \deg v \right) \\
&= p_2 q_1 + p_1 q_2.
\end{aligned}$$

Exercise 1.3. Show that if G is an r-regular graph and H is an s-regular graph, then $G \times H$ is a regular graph of regularity $r + s$.

1.3 Subgraphs

A subgraph of a graph G is a graph H whose vertex and edge sets are, respectively, subsets of the vertex and edge sets of G – i.e. $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. A spanning subgraph of G is a subgraph containing all the vertices of G. A subgraph H of a graph G is an induced subgraph if for all vertices $u, v \in V(H)$, $uv \in E(H)$ whenever $uv \in E(G)$ – i.e. every edge of G that joins two vertices of H is present in H.

Example 1.3. Fig. 1 shows a graph G and three subgraphs H_1, H_2, and H_3 of G. The subgraph H_1 is neither a spanning subgraph (e.g. it does not contain v_3), nor an induced subgraph (e.g. it contains vertices v_2 and v_5 of G but not the edge v_2v_5). H_2 is a spanning subgraph and H_3 is an induced subgraph of G.

A graph is complete if all its vertices are adjacent to one another. The complete graph of order n is denoted by K_n. As there is an edge

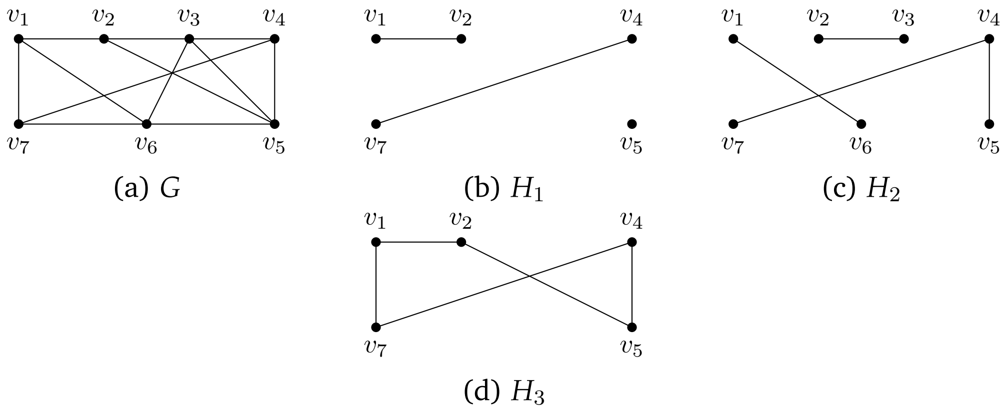


Figure 1: A graph with three of its subgraphs

between every pair of distinct vertices of K_n, the size of K_n is $\binom{n}{2}$. A totally disconnected graph has no edges.

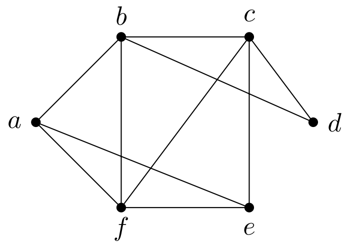
1.4 Walks, Paths, and Cycles

A walk in a simple graph G is a sequence of vertices $v_1, v_2, \ldots, v_k$ such that $v_i \sim v_{i+1}$ for $i = 1, 2, \ldots, k-1$. This is a walk from v_1 to v_k or between v_1 and v_k, having length $k-1$. We may also say that this is a walk of length $k-1$ starting at v_1 and ending at v_k.

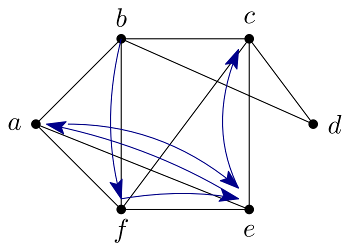
A walk in which no vertex is repeated is a path. Note that the length of a path is the number of edges in it.

A closed walk in G is a walk starting and ending at the same vertex. A closed walk in which no vertex is repeated (except for the start and end vertices being the same) is a cycle. The length of a cycle is also equal to the number of vertices in it. A cycle of the form $v_1, \ldots, v_{k-1}, v_k$ is usually referred to as "the cycle $v_1, \ldots, v_{k-1}$".

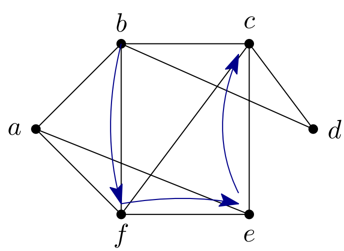
Example 1.4. Consider the graph G shown below.



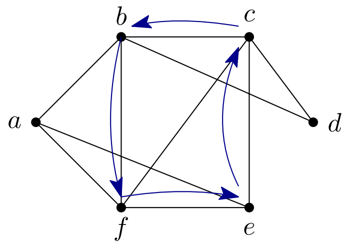
A walk of length 5 from b to c in G is b, f, e, a, e, c.



This walk is not a path, since the vertex e is repeated. An example of a path from b to c in the same graph G is b, f, e, c. The length of this path is 3.



A cycle of length 4 in G is b, f, e, c, b.



Theorem 1.5. *If $\delta(G) \geq 2$, then G contains a path of length $\delta(G)$ and a cycle of length greater than $\delta(G)$.*

Proof. Let $P = v_0 v_1 \cdots v_k$ be a longest path in G. Then all the neighbours of v_0 must be on P (for otherwise, we would get a path longer than P). Hence, $k \geq \deg v_0 \geq \delta(G)$, so P has length at least $\delta(G)$.

Next, let $i \leq k$ be maximal such that $v_i \sim v_0$. Since all $\deg v_0$ neighbours of v_0 occur among $v_1, \ldots, v_i$, we have $i \geq \deg v_0 \geq \delta(G)$. Therefore, $v_0 v_1 \cdots v_i v_0$ is a cycle of length $i + 1 > \delta(G)$. □

Exercise 1.4. Prove that any two longest paths in a connected graph share at least one common vertex.

1.5 Distances and Connectedness

The distance between two vertices u and v of a graph G is the length of a shortest path between u and v – such a shortest path between u and v is a geodesic from u to v. Thus, the distance between u and v in G, denoted by $d_G(u, v)$, is the length of a geodesic from u to v in G. If there is no path from u to v in G, then we set $d_G(u, v) = \infty$. When the graph G is clear from the context, we will write $d_G(u, v)$ as $d(u, v)$.

A graph is connected if there is a path between every two of its vertices. Otherwise, it is disconnected. A (connected) component of a graph G is a maximal connected subgraph of G. Thus, a graph G is connected if and only if it has exactly one component.

The eccentricity of a vertex v, denoted by $\operatorname{ecc}(v)$, is the maximum of all distances from v to any other vertex. That is,

$$\operatorname{ecc}(v) = \max_{u \in V(G)} d(u, v).$$

The diameter of G is the maximum of the eccentricities of vertices of G, and the radius of G is the minimum of the eccentricities of vertices of G. They are denoted by $\operatorname{diam}(G)$ and $\operatorname{rad}(G)$ respectively. Thus,

$$\begin{aligned} \operatorname{diam}(G) &= \max_{v \in V(G)} \operatorname{ecc}(v) = \max_{u,v \in V(G)} d(u, v) \\ \operatorname{rad}(G) &= \min_{v \in V(G)} \operatorname{ecc}(v) = \min_{v \in V(G)} \max_{u \in V(G)} d(u, v). \end{aligned}$$

The set of all vertices of G having minimum eccentricity,i.e. the set of all $v \in V(G)$ such that $\operatorname{ecc} v = \operatorname{rad} G$, is the centre of G.

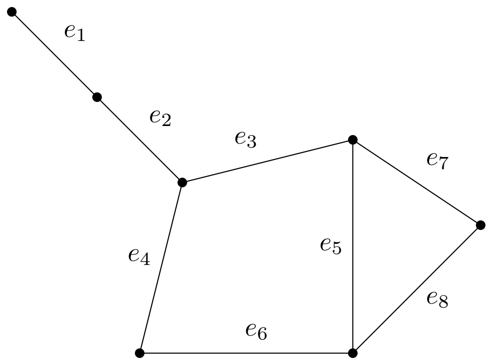
Exercise 1.5. Prove that for any connected graph G, $\operatorname{rad} G \leq \operatorname{diam} G \leq 2 \operatorname{rad} G$.

1.6 Line Graphs

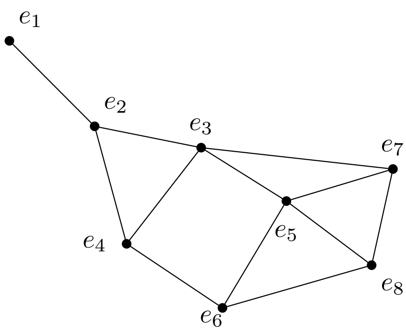
Let G be a graph with at least one edge. The line graph of G is the graph $L(G)$ whose vertex set is the edge set of G, in which two vertices e and f are adjacent if the edges e and f of G are adjacent.

Example 1.6. Fig. 2 shows a graph G with eight edges and and its line graph $L(G)$.

If u and v are edges joined by an edge e in G, then in $L(G)$, the vertex e will be adjacent to all the vertices corresponding to the edges of G other than e that are incident with u and v (see Fig. 3). Thus, $\deg_{L(G)} e = \deg_G u + \deg_G v - 2$.

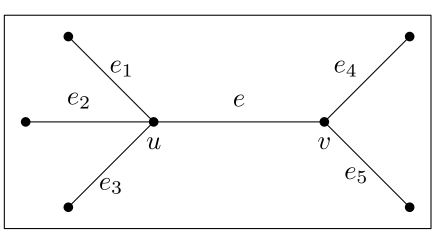


(a) G

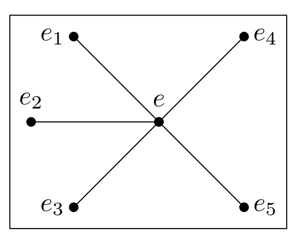


(b) $L(G)$

Figure 2: A graph and its line graph



G



$L(G)$

Figure 3: Neighbourhood of a vertex in $L(G)$

Theorem 1.7. *For any graph G of size $m \geq 1$, its line graph $L(G)$ has order m and size* $\sum_{v \in V(G)} \binom{\deg_G v}{2}$.

Proof. By definition, the vertices of $L(G)$ are the edges of G, and hence the order of $L(G)$ is the size of G, m.

Now, let $e = uv$ be an edge of G. In $L(G)$, e is a vertex, and it is adjacent to another vertex f if and only if f is an edge of G and is adjacent to e,i.e. f is an edge of G incident with one of the end vertices of e, namely u and v. Thus, the degree of the vertex e of $L(G)$ is equal to the total number of edges that are incident with either u or v in G, except for the edge e itself.

Clearly, u is incident with to $\deg_G u - 1$ edges other than e, and v is incident with $\deg_G v - 1$ edges other than e. Thus, the degree of e in $L(G)$ is $\deg_G u - 1 + \deg_G v - 1$.

By Handshaking Lemma, the size of $L(G)$ is half the sum of its vertex degrees, which is

$$\frac{1}{2} \sum_{uv \in E(G)} (\deg_G u - 1 + \deg_G v - 1).$$

As the summation is over the edges of G, each vertex v of G appears exactly $\deg v$ times in the above summation, which can therefore be written as

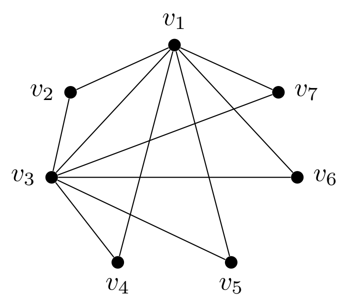
$$\frac{1}{2} \sum_{v \in V(G)} (\deg_G v)(\deg_G v - 1) = \sum_{v \in V(G)} \binom{\deg_G v}{2}.$$ □

1.7 Adjacency Matrices

The adjacency matrix of a graph G of order n, with vertex set $V = \{v_1, \dots, v_n\}$, is the $n \times n$ matrix $A = A(G)$ whose (i, j)-entry is

$$a_{ij} = \begin{cases} 1, & v_i \sim v_j \\ 0, & v_i \nsim v_j. \end{cases}$$

Example 1.8. The adjacency matrix of the graph



is

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Observe that, as the graphs we discuss are simple graphs and therefore have no self-loops on vertices, no vertex is adjacent to itself –i.e. $a_{ii} = 0$ for all $i = 1, \dots, n$. Also, since the graphs are undirected, $v_i \sim v_j$ if and only if $v_j \sim v_i$ –i.e. $a_{ij} = a_{ji}$. Thus, we have the following observation.

Observation 1.9. *The adjacency matrix of a (simple, undirected) graph is a symmetric, zero-diagonal,* $(0,1)$*-matrix.*

In the i^{th} row of the adjacency matrix, for each j, the j^{th} entry is 1 if v_j is adjacent to v_i, and 0 otherwise. That is, the number of 1s in the i^{th} row is the number of vertices adjacent to v_i, or in other words, the degree of v_i. Thus, the row sums of A are the vertex degrees. Observe that $A\mathbb{1}$ is the vector of row sums, where $\mathbb{1}$ is the vector (of suitable size) with all entries equal to 1. For instance, with the matrix A given in Example 1.8,

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 6 \\ 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}.$$

From the Handshaking Lemma and the preceding observation, it follows that the sum of all entries of A is twice the number of edges of the graph.

Theorem 1.10. *Let A be the adjacency matrix of a graph G with vertex set $\{v_1, \ldots, v_n\}$. Then the (i,j)-entry of A^m, for any positive integer m, is the number of walks of length m from v_i to v_j.*

Proof. We prove the result by induction on m. For $m = 1$, the (i,j)-entry of $A^1 = A$ is a_{ij}, which is 1 if and only if v_i is adjacent to v_j,i.e. if and only if there is a walk of length 1 (namely, an edge) from v_i to v_j. Thus, the result holds for $m = 1$.

Now suppose, for the sake of induction, that the result holds for some $m \geq 1$, and consider A^{m+1}. The (i,j)-entry of A^{m+1} is

$$(A^{m+1})_{ij} = \sum_{k=1}^{n} (A^m)_{ik} a_{kj}.$$

First, note that $a_{kj} = 1$ if and only if $v_k \sim v_j$. Therefore, the above sum is equal to the sum of all $(A^m)_{ik}$ where $v_k \sim v_j$. Now, by the induction hypothesis, $(A^m)_{ik}$ is the number of walks of length m from v_i to v_k. If v_k is adjacent to v_j, then each walk of length m from v_i to v_k, together with the edge from v_k to v_j, forms a walk of length $m+1$ from v_i to v_j. Thus, for each k such that $v_k \sim v_j$, $(A^m)_{ik}a_{kj} = (A^m)_{ik}$ is the number of walks of length $m+1$ from v_i to v_j that pass through k. Summing over all k, this gives the total number of walks of length $m+1$ from v_i to v_j. Hence the result follows by induction. □

Theorem 1.11. *Let A be the adjacency matrix of a graph G with vertex set $\{v_1, \ldots, v_n\}$. Then*

(i) $\operatorname{tr}(A^2) = 2|E(G)|$
(ii) $\operatorname{tr}(A^3) = 6c_3(G)$

where $c_3(G)$ denotes the number of triangles in G.

Proof. We know that $(A^m)_{ij}$ is the number of walks of length m from v_i to v_j.

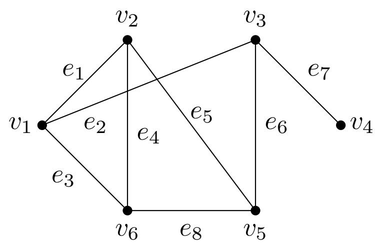
(i) Hence, the i^{th} diagonal entry of A^2, viz. $(A^2)_{ii}$, is the number of walks of length 2 from v_i to itself. Any walk of length 2 from v_i to itself is of the form $v_iv_jv_i$, where v_j is a vertex adjacent to v_i. Thus, the number of such walks is equal to the number of vertices adjacent to v_i,i.e. $\deg v_i$. Hence, $\operatorname{tr}(A^2) = \sum_{i=1}^{n} \deg v_i = 2|E(G)|$, by Handshaking Lemma.

(ii) Similarly, $(A^3)_{ii}$ is the number of walks of length 3 from v_i to itself. Any such walk is of the form $v_iv_jv_kv_i$, which implies that v_i, v_j, and v_k form a triangle in G. Moreover, each such triangle corresponds to two distinct walks from v_i to itself, when traversed in the two opposite directions. Thus, $(A^3)_{ii}$ is twice the number of triangles having v_i as one of its vertices. Therefore, $\operatorname{tr}(A^3)$ is $6c_3(G)$, since each triangle contains three vertices, each of which counts the triangle twice in the sum $\operatorname{tr}(A^3)$. □

1.8 Incidence Matrices

Consider an (n, m)-graph G having vertex set $\{v_1, \dots, v_n\}$ and edge set $\{e_1, \dots, e_m\}$. The incidence matrix of G is the $n \times m$ matrix $B = B(G)$ whose (i, j)-entry is 1 if the vertex v_i is incident with the edge e_j, and 0 otherwise.

Example 1.12. The incidence matrix of the graph



is

$$B = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Each column of the incidence matrix corresponds to an edge of the graph, and the only non-zero entries in the column correspond to the end-vertices of the edge – thus, each column contains exactly two 1s. In a simple graph, there is at most one edge between a given pair of vertices, and hence no two columns of the incidence matrix can be equal. We therefore have the following observation.

Observation 1.13. *The incidence matrix of a (simple, undirected) graph is a $(0, 1)$-matrix in which every column has exactly two 1s, and no two columns are equal.*

Theorem 1.14. *If B is the incidence matrix of a graph G, then the adjacency matrix of its line graph $L(G)$ is*

$$A(L(G)) = B^T B - 2I$$

where I is the identity matrix of order $|E(G)|$.

Proof. Let G be an (n, m)-graph with edge set $E = \{e_1, \ldots, e_m\}$. The incidence matrix $B = B(G)$ is of order $n \times m$, and therefore $B^T B - 2I$ (say) is of order $m \times m$.

Observe that the (i, j) entry of $B^T B$ is the dot product of the i^{th} and j^{th} columns of B. As each column of B has exactly two non-zero entries, both equal to 1, the (i, i) entry of $B^T B$ is 2, and therefore the diagonal entries of $B^T B - 2I$ are all zero.

For $i \neq j$, the dot product of the i^{th} and j^{th} columns of B will be 1 if the edges e_i and e_j are adjacent (i.e. have one vertex in common), and 0 otherwise – note that the dot product cannot be 2, as no two distinct columns of B can be equal. Thus, for $i \neq j$, the (i, j)-entry of $B^T B$, which is equal to the (i, j) entry of $B^T B - 2I$, is 1 if and only if the edges e_i and e_j are adjacent in G, or equivalently, the vertices e_i and e_j of $L(G)$ are adjacent. Thus, $B^T B - 2I$ is the adjacency matrix of $L(G)$. □

1.9 Application: Centrality and Clustering

A social network can be modelled by a graph whose vertices represent people and whose edges represent relationships or interactions. Different numerical measures capture different meanings of “importance”, so identifying an influencer depends on the purpose of the analysis.

1.9.1 Centrality

Let G be a connected graph of order $n \geq 2$. The following are common measures of the centrality of a vertex v.

(i) The degree centrality

$$C_D(v) = \frac{\deg(v)}{n-1}$$

measures the proportion of the network directly connected to v. A large value suggests high immediate reach.

(ii) The closeness centrality

$$C_C(v) = \frac{n-1}{\displaystyle\sum_{\substack{u \in V(G) \\ u \neq v}} d(u,v)}$$

is large when v is only a few steps away from the rest of the network. Such a vertex can spread information efficiently.

(iii) For distinct vertices s, t, let σ_{st} be the number of shortest s–t paths and let $\sigma_{st}(v)$ be the number of these paths that pass through v. The betweenness centrality

$$C_B(v) = \sum_{\{s,t\} \subseteq V(G) - \{v\}} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

measures how often v acts as a bridge between other vertices. It is particularly useful for finding brokers or potential bottlenecks.

(iv) An eigenvector centrality vector is a non-negative vector $\mathbf{x}$ satisfying

$$A\mathbf{x} = \lambda\mathbf{x},$$

where A is the adjacency matrix and λ is its largest eigenvalue. Its entry x_v assigns a high score to v when v is adjacent to other high-scoring vertices.

Remark. No single centrality measure always identifies the "most influential" vertex. Degree measures direct popularity, closeness measures rapid access, betweenness measures brokerage, and eigenvector centrality values connections to already important vertices.

1.9.2 Clustering

For a vertex v, let $N(v)$ be its set of neighbours and let $e(N(v))$ be the number of edges having both end-vertices in $N(v)$. The local clustering coefficient of v is

$$C(v) = \begin{cases} \dfrac{e(N(v))}{\binom{\deg(v)}{2}}, & \deg(v) \geq 2, \\ 0, & \deg(v) < 2. \end{cases}$$

It is the proportion of pairs of neighbours of v that are themselves adjacent. The average clustering coefficient of the network is

$$\overline{C}(G) = \frac{1}{n} \sum_{v \in V(G)} C(v).$$

A large local coefficient suggests that v belongs to a tightly connected community. A vertex with high centrality but relatively low clustering may instead connect different communities.

Exercise 1.6. For the graph in Example 1.8, compute the degree and closeness centralities and the local clustering coefficient of every vertex. Which vertices would be selected as influencers by these measures?

Exercise 1.7. Explain how the entries of A^2 and A^3 can help detect common neighbours and triangles in a social network.